

* Unitary operator $U(\Lambda)$ s.t.

$$U(\Lambda)^{-1} \psi(x) U(\Lambda) = \psi(\Lambda^{-1}x)$$

which implies

$$U(\Lambda)^{-1} \partial^\mu \psi(x) U(\Lambda) = \Lambda^\mu_\rho \bar{\partial}^\rho \psi(\Lambda^{-1}x)$$

$$\bar{x} = \Lambda^{-1}x \Rightarrow \frac{\partial}{\partial x^\mu} = \frac{\partial \bar{x}^\sigma}{\partial x^\mu} \frac{\partial}{\partial \bar{x}^\sigma} = \Lambda^{-1\sigma}_\mu \bar{\partial}_\sigma$$

$$\bar{\partial}_\sigma = \Lambda^\lambda_\sigma \partial_\lambda$$

* The above implies we can define a Vector field $A^\mu(x)$ s.t.

$$U(\Lambda)^{-1} A^\mu(x) U(\Lambda) = \Lambda^\mu_\rho A^\rho(\Lambda^{-1}x)$$

Tensor field: $B^{\mu\nu}(x)$

$$U(\Lambda)^{-1} B^{\mu\nu}(x) U(\Lambda) = \Lambda^\mu_\sigma \Lambda^\nu_\lambda B^{\sigma\lambda}(\Lambda^{-1}x)$$

B is $B^{(MN)}$ or $B^{[MN]}$

* For the trace $T(x) \equiv g_{\mu\nu} B^{\mu\nu}(x)$

$$\text{Using } g_{\mu\nu} \Lambda^\mu_\sigma \Lambda^\nu_\lambda = g_{\sigma\lambda}$$

$$g_{\mu\nu} [U(\Lambda)^{-1} B^{\mu\nu} U(\Lambda)] = \Lambda^\mu_\sigma \Lambda^\nu_\lambda B^{\sigma\lambda}(\Lambda^{-1}x)$$

$$U(\Lambda)^{-1} T(x) U(\Lambda) = T(\Lambda^{-1}x)$$

* We can decompose $B^{\mu\nu}(x)$ into its symm + trace + anti-symm.

$$B^{\mu\nu}(x) = A^{\mu\nu}(x) + S^{\mu\nu} + \frac{1}{4} g^{\mu\nu} T(x)$$

$$S^{\mu\nu} g_{\mu\nu} = 0$$

Under transformations these objects do not mix.

$A^{\mu\nu}$

$S^{\mu\nu}$

under $SU(N)$:

$\square$

$\square$

* Consider a field $\varphi_A(x)$: A : Lorentz group index

$$U(\Lambda)^{-1} \varphi_A(x) U(\Lambda) = \rho_A^B(\Lambda) \varphi_B(\Lambda^{-1}x)$$

$\rho_A^B(\Lambda)$: matrix that depends on Λ and obeys

$$\rho_A^B(\Lambda') \rho_B^C(\Lambda) = \rho_A^C(\Lambda'\Lambda)$$

$$\rho: ISO^+(1,3) \rightarrow GL_{10}(\mathbb{C})$$

• For an infinitesimal transformation

$$\Lambda_N^M = \delta_N^M + \delta\omega^M_N$$

$$U(\delta + \delta\omega)_N^M = \delta_N^M + \delta\omega_{\sigma\lambda} \frac{\delta}{\delta\omega^{\sigma\lambda}} U_N^M$$

with

$$\partial/\partial\omega^{\sigma\lambda} U = M_{\sigma\lambda} \frac{i}{2}$$

$$U(\delta + \delta\omega) = \mathbb{I}_{mn} + \frac{i}{2} \delta\omega_{mn} M^{mn}$$

M^{mn} : Generators of the l.a. of $ISO^+(1,3)$

Since

$$U(\Lambda)^{-1} M_{mn} U(\Lambda) = \Lambda_m^\sigma \Lambda_n^\lambda M_{\sigma\lambda}$$

$$\text{Let } \Lambda_N^M = \delta_N^M + \delta\omega^M_N$$

$$\left(\mathbb{I} - \frac{i}{2} \delta\omega^{\sigma\lambda} M_{\sigma\lambda} \right) M_{mn} \left(\mathbb{I} + \frac{i}{2} \delta\omega^{\alpha\beta} M_{\alpha\beta} \right)$$

$$\left(M_{mn} - \frac{i}{2} \delta\omega^{\sigma\lambda} M_{\sigma\lambda} M_{mn} \right) \left(\mathbb{I} + \delta M_{\alpha\beta} \right)$$

$$M_{mn} - \frac{i}{2} \delta\omega^{\sigma\lambda} M_{\sigma\lambda} M_{mn} + \frac{i}{2} M_{mn} \delta\omega^{\alpha\beta} M_{\alpha\beta}$$

$$\begin{aligned} \text{(RHS)} & (\delta_\sigma^m + \delta\omega_\sigma^m) (\delta_\lambda^n + \delta\omega_\lambda^n) M^{\lambda\sigma} \\ &= (M^{\lambda m} + \delta\omega_\sigma^m M^{\lambda\sigma}) (\delta_\lambda^n + \delta\omega_\lambda^n) \\ &= M^{mn} + \delta\omega_\sigma^m M^{\lambda\sigma} + M^{\lambda m} \delta\omega_\lambda^n \end{aligned}$$

$$\begin{aligned} & \hookrightarrow M_{mn} + \delta\omega_\sigma^m M^{\lambda\sigma} + M_{\lambda m} \delta\omega_\nu^\lambda \\ i/2 \delta\omega^{\sigma\lambda} [M_{mn}, M_{\sigma\lambda}] &= \delta\omega^\sigma_\mu M^{\nu\sigma} + M_{\lambda m} \delta\omega_\nu^\lambda \end{aligned}$$

$$\text{(RHS)} = \delta\omega^{\sigma\lambda} (g_{\lambda m} M_{\nu\sigma} + M_{\lambda m} g_{\nu\sigma})$$

$$\therefore \frac{i}{2} [M_{mn}, M_{\sigma\lambda}] = g_{m\lambda} M_{\nu\sigma} - g_{n\sigma} M_{m\lambda}$$

$$\boxed{[M^{\mu\nu}, M^{\rho\sigma}] = i(g^{\mu\rho} M^{\nu\sigma} - (m \leftrightarrow \nu) - (p \leftrightarrow \sigma))}$$

* Extra terms come from $M_{mn}(\Lambda^{-1}x)$

Being expanded

* The Lorentz generators is (Boosts) $\oplus$ (Rotations)

Angular Momenta: $\vec{J} = (J_1, J_2, J_3)$

$$J_i \equiv \frac{1}{2} \epsilon_{ijk} M^{jk}$$

$$*: \Lambda^p \rightarrow \Lambda^{p-p}$$

$$D=4, \quad *: \Lambda^p \rightarrow \Lambda^{4-p}$$

So J_i is the dual of some 3-form

Boosts: $\vec{K} = (K_1, K_2, K_3)$

$$K_i \equiv M^{i0}$$

from which we find

$$[J_i, J_j] = +i \epsilon_{ijk} J_k$$

$$[J_i, K_j] = +i \epsilon_{ijk} K_k$$

$$[K_i, K_j] = -i \epsilon_{ijk} J_k$$

$$\rho_A^B(1+\delta\omega) = \delta_A^B + \frac{i}{2} \delta\omega_{mn} (S^{mn})_A^B$$

$$* \text{ Then } U(\Lambda)^{-1} \varphi_A(x) U(\Lambda) = \rho_A^B \varphi_B(\Lambda^{-1}x)$$

$$\left(\mathbb{1} - \frac{i}{2} \delta\omega^{\alpha\beta} M_{\alpha\beta} \right) \varphi_A(x) \left(\mathbb{1} + \frac{i}{2} \delta\omega^{\lambda\sigma} M_{\lambda\sigma} \right)$$

$$\left(\varphi_A - \frac{i}{2} \delta\omega^{\alpha\beta} M_{\alpha\beta} \varphi_A(x) \right) \left(\mathbb{1} + \frac{i}{2} \right)$$

$$\varphi_A - \frac{i}{2} \delta\omega^{\alpha\beta} M_{\alpha\beta} \varphi_A(x) + \frac{i}{2} \varphi_A(x) \delta\omega^{\lambda\sigma} M_{\lambda\sigma}$$

$$(i/2) \delta\omega^{\lambda\sigma} [\varphi_A(x), M_{\lambda\sigma}]$$

$$(\text{RHS}) \left(\delta_A^B + \frac{i}{2} \delta\omega_{\lambda\sigma} (S^{\lambda\sigma})_A^B \right) \left(\varphi_B(x) - \delta\omega_{\lambda\sigma} x^\lambda \partial^\sigma \varphi_B \right)$$

$$\varphi_A + \frac{i}{2} \delta\omega_{\lambda\sigma} (S^{\lambda\sigma})_A^B \varphi_B(x) - \delta\omega_{\lambda\sigma} x^\lambda \partial^\sigma \varphi_A$$

$$\varphi_A + \frac{i}{2} \delta\omega_{\lambda\sigma} (S^{\lambda\sigma})_A^B \varphi_B(x) - \frac{i}{2} \delta\omega_{\lambda\sigma} [x^\lambda \partial^\sigma - x^\sigma \partial^\lambda] \varphi_A$$

$$\therefore \frac{i}{2} [\varphi_A(x), M_{\lambda\sigma}] = \frac{i}{2} (S_{\lambda\sigma})_A^B \varphi_B(x) - \frac{i}{2} (x_\lambda \partial_\sigma - x_\sigma \partial_\lambda) \varphi_A$$

$$[\varphi_A(x), M^{\mu\nu}] = (S^{\mu\nu})_A^B \varphi_B(x) + \mathcal{L}^{\mu\nu} \varphi_A(x)$$

$$\mathcal{L}^{\mu\nu} \equiv \frac{1}{i} (x^\mu \partial^\nu - x^\nu \partial^\mu)$$

For $J_i = \frac{1}{2} \varepsilon_{ijk} M^{jk}$, we can find $(2j+1) \times (2j+1)$ matrices J_1, J_2, J_3 with J_3 having eigenvalues $-j, \dots, j$
 $j = 0, \frac{1}{2}, 1, \dots$

$$G = SO(3) \ni g = e^{\lambda T}$$

$$\mathfrak{g} = SU(2) \ni \tau$$

* N has no physical significance

$$N_i \equiv \frac{1}{2} (J_i - i K_i)$$

$$N_i^\dagger \equiv \frac{1}{2} (J_i + i K_i)$$

and so

$$[N_i, N_j] = i \varepsilon_{ijk} N_k$$

$$[N_i^\dagger, N_j^\dagger] = i \varepsilon_{ijk} N_k^\dagger$$

$$[N_i, N_j^\dagger] = 0$$

• Thus, there are two inequivalent reps of $SU(2)$ which are related via herm. conjugation.

Thus, the reps of $SU(2)$ in $d=4$ is determined via two values (n, n') which $n \in \frac{1}{2}\mathbb{Z} + \mathbb{Z}$

$$(2n+1, 2n'+1)$$

$$\mathfrak{su}(2) \oplus \overline{\mathfrak{su}(2)}$$

* The angular momentum of a field

$$J_i = N_i + N_i^*$$

So the allowed values of its angular momentum j given n and n'

$$j \in \{ |n-n'|, |n-n'|+1, \dots, n+n' \}$$

$(1,1)$: Scalar (Singlet)

$(2,1)$: Left handed spinor

$(1,2)$: Right handed spinor

$(2,2)$: Vector

$(0,0)$

$(\frac{1}{2}, 0)$

$(0, \frac{1}{2})$

$(\frac{1}{2}, \frac{1}{2})$

ex: Recall j can be $|n-n'|, \dots, n+n' = j$

so for the vector $(2,2) = (\frac{1}{2}, \frac{1}{2})$

$$j = \frac{1}{2} - \frac{1}{2} = 0$$

$$j = 1$$

$$\Rightarrow j = \begin{cases} 0, & x^0 \\ 1, & x^i \end{cases}$$